Extend the Windows Server 2022 System Volume by Deleting and Recreating the Recovery Partition

Validated workflow: Uses `mountvol R: /D` to remove the temporary drive letter and includes a final reboot verification.

Purpose

This procedure describes how to extend the Windows Server 2022 system volume when the Recovery partition is located between the C: drive and unallocated space. The process preserves the Windows Recovery Environment (WinRE), extends C:, recreates the Recovery partition, re-enables WinRE, removes the temporary drive letter, and verifies the configuration after reboot.

Prerequisites

• Windows Server 2022.

• GPT-partitioned system disk.

• The VMware virtual disk has already been expanded.

• Disk layout resembles: `[C:] [Recovery] [Unallocated]`.

• Administrator Command Prompt.

• Windows Server 2022 installation ISO, only if `winre.wim` must be extracted.

Recommendation: Take a VMware snapshot or ensure a valid backup exists before modifying partitions.

Important: Confirm the correct disk and partition numbers before running any delete command. The examples use Disk 0, but your server may differ.

Step 1 - Verify the Current Recovery Environment

```
reagentc /info

diskpart
list disk
select disk 0
list partition
exit
```

Record the current Recovery partition number and confirm that the unallocated space is located after it.

Step 2 - Preserve the WinRE Image

Option A - winre.wim exists locally

Check whether the file exists:

```
dir /a C:\Windows\System32\Recovery
```

If `winre.wim` is present, back it up:

```
md C:\WinRE-Backup
copy C:\Windows\System32\Recovery\winre.wim C:\WinRE-Backup\
dir /a C:\WinRE-Backup
```

Option B - winre.wim is missing: extract it from the Server 2022 ISO

1. Mount the Windows Server 2022 ISO. In these examples the mounted ISO is drive D:. Change the drive letter if necessary.

2. Verify whether the ISO contains `install.wim` or `install.esd`:

```
dir D:\sources\install.*
```

3. Determine the installed Windows Server edition:

```
dism /online /Get-CurrentEdition
systeminfo | findstr /B /C:"OS Name"
```

4. List the indexes available in the installation image. Run the command that matches the file on the ISO:

```
dism /Get-WimInfo /WimFile:D:\sources\install.wim
```

or:

```
dism /Get-WimInfo /WimFile:D:\sources\install.esd
```

Identify the index whose Name and Description match the installed edition, for example Windows Server 2022 Standard (Desktop Experience) or Windows Server 2022 Datacenter (Desktop Experience). Do not assume the index number; it can vary by ISO.

5. Create the mount directory:

```
md C:\Mount
```

6. Check for stale or incomplete DISM mounts:

```
dism /Get-MountedWimInfo
```

If a stale mount exists, clean it up before continuing:

```
dism /Cleanup-Wim
```

7. Mount the correct image index as read-only.

For install.wim:

```
dism /Mount-Wim /WimFile:D:\sources\install.wim /Index:<IndexNumber> /MountDir:C:\Mount /ReadOnly
```

For install.esd:

```
dism /Mount-Image /ImageFile:D:\sources\install.esd /Index:<IndexNumber> /MountDir:C:\Mount /ReadOnly
```

Replace `<IndexNumber>` with the index identified in the previous step.

8. Verify that the mounted image contains the WinRE file:

```
dir /a C:\Mount\Windows\System32\Recovery
```

9. Copy `winre.wim` to the backup folder:

```
md C:\WinRE-Backup
copy C:\Mount\Windows\System32\Recovery\winre.wim C:\WinRE-Backup\
dir /a C:\WinRE-Backup
```

10. Unmount the installation image:

```
dism /Unmount-Wim /MountDir:C:\Mount /Discard
```

Verify the mount directory is no longer active:

```
dism /Get-MountedWimInfo
```

Step 3 - Disable WinRE

```
reagentc /disable
reagentc /info
```

Expected result: **Windows RE status: Disabled**.

Step 4 - Delete the Existing Recovery Partition

```
diskpart
select disk 0
list partition
select partition <RecoveryPartitionNumber>
detail partition
delete partition override
list partition
exit
```

> Confirm that the selected partition is the existing Recovery partition before running `delete partition override`.

Step 5 - Extend the C: Drive

```
diskpart
select disk 0
list volume
select volume C
extend
exit
```

Alternatively, use Disk Management: right-click C: and select **Extend Volume**.

Leave at least 1024 MB available at the end of the disk for the new Recovery partition. If you extended C: into all available space, shrink C: by 1024 MB before creating the Recovery partition.

Step 6 - Recreate the Recovery Partition

If required, shrink C: by 1024 MB in Disk Management or DiskPart. Then create the new Recovery partition:

```
diskpart
select disk 0
create partition primary size=1024
format quick fs=ntfs label="Windows RE"
assign letter=R
list partition
select partition <NewPartitionNumber>
set id=de94bba4-06d1-4d40-a16a-bfd50179d6ac
gpt attributes=0x8000000000000001
detail partition
exit
```

Expected partition properties:

```
Type     : de94bba4-06d1-4d40-a16a-bfd50179d6ac
Hidden   : Yes
Required : Yes
Attrib   : 0x8000000000000001
Fs       : NTFS
Size     : 1024 MB
```

Step 7 - Restore the WinRE Files

```
md R:\Recovery
md R:\Recovery\WindowsRE
copy C:\WinRE-Backup\winre.wim R:\Recovery\WindowsRE\
dir /a R:\Recovery\WindowsRE
```

The directory must contain `winre.wim`. `boot.sdi` and `ReAgent.xml` may also be present or created during registration.

Step 8 - Register and Enable WinRE

```
reagentc /setreimage /path R:\Recovery\WindowsRE
```

```
reagentc /enable
reagentc /info
```

Expected output:

```
Windows RE status: Enabled
Windows RE location:
\\?\GLOBALROOT\device\harddisk0\partitionX\Recovery\WindowsRE
```

> Confirm WinRE is enabled and points to the newly created Recovery partition before removing the temporary drive letter.

Step 9 - Remove the Temporary Drive Letter

```
mountvol R: /D
```

This removes the R: mount point without using DiskPart.

Step 10 - Verify the Recovery Partition

```
diskpart
select disk 0
list partition
select partition <RecoveryPartitionNumber>
detail partition
exit
```

Confirm:

- Type is `de94bba4-06d1-4d40-a16a-bfd50179d6ac`.
- Hidden is Yes.
- Required is Yes.
- Attrib is `0x8000000000000001`.
- Filesystem is NTFS.
- Size is approximately 1024 MB.
- No drive letter is assigned.
- Status is Healthy.

Step 11 - Reboot and Perform Final Verification

Reboot the server. After the reboot, open an Administrator Command Prompt and run:

```
reagentc /info
```

Expected result:

```
Windows RE status: Enabled
Windows RE location:
\\?\GLOBALROOT\device\harddisk0\partitionX\Recovery\WindowsRE
```

Also confirm that the C: drive reflects the newly extended size.

Completion Checklist

- C: has been successfully extended.
- A 1024 MB NTFS Recovery partition exists at the end of the disk.
- The Recovery partition has the correct GPT type GUID.
- The Recovery partition is Hidden and Required.

- The Recovery partition has no drive letter.
- WinRE remains Enabled after reboot.
- WinRE points to the recreated Recovery partition.